

JAYANT KUMAR

AI/ML Engineer | Predictive Analytics & Generative AI

 kjayant8628@gmail.com

 [+91-9711989002](tel:+91-9711989002)

 [Linkedin](#)

 [Github](#)

Work Experience

AI/ML & Data Analytics Intern, Siemens Energy Python, Azure OpenAI, GPT APIs, Pydantic, JSON Pipelines, Azure Functions, Power BI, Microsoft Copilot - Engineered an AI-powered bid automation platform that reduced RFP processing time from 16–40 hours of manual effort to under 3 hours, accelerating technical offer generation across 20+ RFPs/month. - Developed a hybrid NLP + LLM reasoning engine for requirement classification, achieving 90%+ deterministic compliance accuracy and replacing error-prone manual review workflows. - Built document preprocessing pipelines for parsing, chunking, and discrepancy detection, automating generation of 50+ structured DOCX/JSON/XLSX outputs per month and eliminating manual formatting overhead. - Implemented schema-validated, section-level change logs enabling 100% traceable audit trails, reducing review cycles and eliminating ambiguity in bid preparation workflows.	01/2026 – 06/2026 Gurgaon
AI & Backend Developer Intern, HCL Technologies Python, FastAPI, PostgreSQL, REST APIs, Async Processing, Workflow Optimization - Designed and deployed FastAPI-based backend services supporting 30+ concurrent users at 99%+ availability, enabling reliable AI-driven workflows across production and testing environments. - Implemented JWT authentication and Role-Based Access Control (RBAC), eliminating unauthorized access pathways and measurably reducing the application's attack surface. - Improved backend response efficiency by 20–40% through asynchronous request handling, reducing latency for time-sensitive AI workflow operations.	05/2025 – 07/2025 Noida
Data Analyst Intern, HCL Technologies Python, Pandas, Anomaly Detection, Operational Analytics, Security Monitoring - Analyzed tens of thousands of enterprise user activity and operational records to detect unauthorized usage patterns, identifying and remediating 15+ Shadow IT instances across critical systems. - Applied anomaly detection techniques to reduce manual log investigation from several hours to targeted, near-instant flagging, contributing to an estimated 40% reduction in potential data breach risk.	06/2024 – 08/2024 Noida

Academic Projects

Predictive Drug–Drug Interaction Analysis using Graph Neural Networks & Knowledge Graphs Python, PyTorch, PyTorch Geometric, RDKit, PyKEEN, Pandas, NumPy, Scikit-learn - Developed a hybrid GNN-Knowledge Graph architecture for Drug–Drug Interaction prediction on 2.9M+ biomedical drug-pair records, outperforming structure-only and knowledge-graph-only baselines. - Designed end-to-end data preprocessing, feature engineering, model training, and inference pipelines integrating DrugBank, KEGG, UniProt, ATC, and MeSH datasets with graph-based molecular representations. - Achieved 92.3% accuracy and 0.932 AUROC using message-passing neural networks, embedding fusion, and hyperparameter optimization for multimodal predictive modeling.
Enterprise Multi-Agent RAG System for Intelligent Document Search Python, LangChain, LangGraph, FastAPI, OpenAI, Vector Database, Embeddings, NLP - Architected a multi-agent RAG platform using LangGraph, LLMs, and vector embeddings, implementing NLP pipelines for document parsing, chunking, semantic search, and question answering across unstructured data. - Optimized retrieval orchestration and prompt-engineering workflows to improve contextual relevance, reduce hallucinations, and accelerate information retrieval from several minutes of manual search to under 5 seconds. - Developed FastAPI-based services for scalable document intelligence, indexing and retrieving thousands of enterprise documents to enhance knowledge discovery and reduce redundant manual review.
Indian Tourism Demand Forecasting and Trend Analysis Python, ARIMA, XGBoost, Pandas, Numpy - Trained and benchmarked ARIMA and XGBoost models on multi-source tourism datasets, generating 3–12 month demand forecasts and demonstrating XGBoost's superior resilience to non-linear disruptions like COVID-19. - Delivered 15+ scenario-driven analytical insights for strategic planning by evaluating models across RMSE, MAE, and MAPE metrics on volatile, real-world time-series data.

Skills

Languages Python, SQL	Machine Learning & AI LangChain, LangGraph, RAG, Azure OpenAI, Azure Document Intelligence, Prompt Engineering, Embeddings, Vector Databases	Data & Analytics Power BI, Tableau, Pandas, NumPy, Matplotlib, Seaborn, Pytorch	Platforms Git, GitHub, Azure Functions, Jupyter Notebook
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Educational Qualification

Bachelor of Technology in Computer Science, Shiv Nadar University	08/2022 – 05/2026 Uttar Pradesh
12th CBSE Board, Indraprastha Public School Grade: 80%	04/2021 – 05/2022 New Delhi
10th CBSE Board, Delhi Public School Grade: 92%	04/2019 – 05/2020 Gurugram